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17 April 1987
SUPERSEDING
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26 March 1979

MILITARY SPECIFICATION

CEREAL BARS, SURVIVAL-TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers survival-type cereal bars for use by the Armed Forces as a component of operational rations.

1.2 Classification. The bars shall be of the following types, as specified (see 6.1):

Type I	-	Corn flake bar
Type II	-	Rice-corn flake bar
Type III	-	Fruit flavored corn flake bar
Class 1	-	Orange flavor
Class 2	-	Lemon flavor
Type IV	-	Granola Bar
Style A	-	2-ounce (57 g) bar with rounded corners
Style B	-	1-1/2 ounce (43 g) bar

Fortified
Nonfortified

2. APPLICABLE DOCUMENTS

2.1 Government document

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8920

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2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-C-110	-	Cellophane (Coated and Noncoated Regenerated Cellulose Film)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

MILITARY

MIL-L-10547	-	Liner, Case and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence Items, General Specification for

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection and Grading of Eggs and Egg Products

(Application for copies should be addressed to the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, Inc., 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

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(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth street, Champaign, IL 61820.)

ASSOCIATION OF VITAMIN CHEMISTS, INC.

Methods of Vitamin Assay

(Application for copies should be addressed to the Interscience Publishers, a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Corn flakes, plain. Plain corn flakes shall be prepared from clean, sound corn (not high protein or lysine corn). The corn shall be scoured, degermed, dehulled, and processed, with added flavoring agents, into flakes that are reasonably uniform in size. The flakes shall be tender, crisp, and of good flavor.

3.2.2 Corn flakes, sugar-coated. Sugar-coated corn flakes shall conform in all respects to 3.2.1, except that the flakes shall be coated with sugar by applying a concentrated sugar solution to the flakes and evaporating the excess water.

3.2.3 Rice, oven-puffed. Crisp, oven-puffed rice shall be prepared from cleaned, sound rice. The rice material shall be processed with sugar and malt flavoring, prior to oven-puffing.

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3.2.4 Shortening. The shortening shall be derived from refined hydrogenated vegetable oils and shall have a stability of not less than 100 hours prior to the addition of antioxidant, as determined by the active oxygen method (AOM) (see 4.6.2). The shortening shall contain antioxidants with active ingredients of TBHQ, BHA, BHT, propyl gallate, alone or in combination, in accordance with commercial practice, at the maximum allowable level.

3.2.5 Egg whites, dried. The dried egg whites shall contain no whipping aids, and shall be processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59) as evidenced by the USDA egg products inspection shield on the label. Each lot of dried egg whites shall be sampled and analyzed for salmonella by the USDA as evidenced by a USDA Egg Products Inspection Certificate.

3.2.6 Starch, tapioca, modified, pregelatinized, instant. The starch shall be a white, odorless, finely ground powder. It shall be modified, pregelatinized, crosslinked, and stabilized. The starch shall have a maximum moisture content of 8.0 percent and pH of 5.0 to 7.0.

3.2.7 Sorbitol. Sorbitol shall be liquid containing minimum 70 percent sorbitol conforming to the Food Chemicals Codex.

3.2.8 Water. Water used for formulation and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.9 Fruit flavoring. Orange flavor (for class 1 bars) or lemon flavor (for class 2 bars) shall be cold pressed or concentrated citrus oils which have been entrapped in a matrix containing corn syrup solids, glycerol, and other ingredients as required. The particles of the flavoring material shall be coated or treated in a manner such that their hygroscopicity is greatly reduced. The flavoring material shall result in a finished product possessing a characteristic flavor and odor.

3.2.10 Vitamins. Vitamin A shall be dry, water dispersible vitamin A palmitate stabilized in gelatin, gums, or other edible materials with or without sugar. Ascorbic acid (vitamin C), thiamine hydrochloride, and pyridoxine hydrochloride shall be of Food Chemicals Codex grade. In addition, the particle size of all vitamins shall be such that the vitamins will be uniformly distributed throughout the bar.

3.2.11 Rolled oats. The rolled oats shall be thin, uniform flakes produced by rolling or pressing whole oat groats. The oats shall be heat treated to destroy enzyme activity. The oats shall test negative to tyrosinase (catechol oxidase, see 4.6.4).

3.2.12 Coconut, dried, sweetened, flaked. Coconut shall be salmonellae-free and of the best quality flaked type, sweetened. It shall possess a characteristic coconut odor and flavor without off-odor and flavor. It shall contain not more than 11.0 percent moisture.

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3.2.13 Coconut fat. Coconut fat shall be a refined coconut vegetable shortening and shall conform to the following requirements:

Flavor and Odor:	Shall have coconut oil characteristics.
Moisture and Impurities:	0.10 percent maximum.
Wiley Melting Point:	73° to 102°F.
Solid Fat Index:	50°F 56-66, 70F° 25-40, 80°F 1-15, 92°F 0-7, 100°F 5 maximum.
Peroxide Value:	0.5 milliequivalent of peroxide per kg of fat.
Free Fatty Acids:	0.05 percent maximum.
Stability (AOM):	100 hours minimum.
Iodine Value:	7.0 maximum.
Color (Lovibond):	1.5R/15.0Y maximum.

3.2.14 Honey. The honey shall be high commercial grade, and be free of any off flavor or off-odor.

3.2.15 Glycerol. Glycerol shall comply with the Food Chemicals Codex.

3.2.16 Vanilla flavoring. Vanilla flavoring shall be single strength pure vanilla extract. Multiple fold pure vanilla extract may be substituted for single strength, provided appropriate percent formula adjustments are made.

3.2.17 Rice, crisped. Crisped rice shall be processed from clean, sound rice and have a density between 100-160 g/L (standard confectionary or full grain) and will typically contain 7.5 $\pm$ 2 gram sucrose per 100 grams.

3.2.18 Cinnamon, oleoresin, encapsulated. Encapsulated cinnamon oleoresin shall be obtained by the approved solvent extraction of the dried bark of the genus cinamomum (Korintji type) and shall be a powder obtained by spray drying the extract on a dextrin carrier. The product shall have a typical four-fold strength vs. ground cinnamon, a minimum volatile oil content of 8.7 percent, and a maximum moisture of 8.0 percent.

3.2.19 Brown sugar. Sugar shall be of typical flavor and uniform light to medium brown color. It shall possess a characteristic brown sugar flavor.

3.3 Formulas. The bars (fortified or nonfortified) shall be made from the following ingredients in the proportion shown. In addition, the fortified product shall contain the following per ounce of product: Thiamine hydrochloride -- 0.56 mg; Vitamin A -- not less than 1670 I.U., Ascorbic acid -- not less than 25 mg; Pyridoxine hydrochloride -- not less than 0.67 mg.

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Corn flake, rice-corn flake, and flavored bars

Ingredients	Percent by weight Type I	Type II	Type III
Corn flakes, plain	30	15	0
Corn flakes, sugar-coated	35	35	71
Rice, oven puffed	0	15	0
Shortening, hydrogenated	18	18	18
Egg whites, dried	7	7	7
Starch, instant	4	4	0
Sorbitol solution	5	5	0
Water (maximum)	1	1	2
Fruit flavoring	0	0	as required

Granola bar (type IV)

Ingredients	Percent by weight
Rolled oats	45.0
Coconut, dried, sweetened, flaked	6.5
Coconut fat	10.0
Honey	4.0
Glycerol	3.0
Vanilla flavoring	0.3
Rice, crisped	11.0
Cinnamon, oleoresin	0.2
Brown sugar	12.0
Water	8.0

3.4 Preparation. For types I and II, a paste shall be prepared by blending together the shortening, dried egg whites, and starch and mixing until homogeneous. For type III, the shortening, dried egg whites, and fruit flavor shall be blended together and mixed into a homogeneous paste. In each of these three types, the cereals shall be mixed until the flakes are reduced to a size that will result in a bar of the required weight when compressed to the dimensions specified in 3.4.1. The paste mixture shall be added, in small portions, to the cereal and mixed until uniformly distributed. When vitamin fortification is required (see 6.1), vitamins shall be incorporated in the paste for types I, II, and III and after toasting for type IV at the level specified in 3.3. The water and sorbitol (in the case of types I and II) and the water (in the case of type III) shall be sprinkled over the cereal-paste mixture and the mixing continued for a sufficient time to distribute the solution. The prepared mixture shall be immediately formed into bars in accordance with 3.4.1. For type IV, the coconut fat shall be melted and mixed with honey, vanilla, brown sugar and water. The remaining ingredients shall be added and the product mixed until homogeneous.

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The product shall be toasted until it reaches a temperature of 220°F, then removed from the oven immediately and cooled. Product may be held in accordance with good commercial practice for up to 30 days before forming into bars.

3.4.1 Compression. For style A, the premixed cereal shall be compressed into a bar having the following dimensions: Length 3-1/2 inches, width 1-13/16 inches, thickness 9/16 inch, ± 1/16 inch in any dimension. The corners of each bar shall be rounded. The net weight of the bar shall be not less than 2.0 ounces. For style B, the premixed cereal shall be compressed into a bar having the following dimensions: Length 3-1/8 inches ± 1/16 inch; width 1-1/8 inches ± 1/16 inch; thickness, not more than 3/4 inch for types I, II and III and not more than 1.0 inch for type IV. The net weight of the bar shall be not less than 1.5 ounces. Style A and B bars shall be sufficiently firm to withstand packaging, without crumbling, in accordance with 5.1. Type IV shall be spray conditioned before compression with a 50/50 brown sugar and water solution. No heating shall be used to aid compression or redry bars after compression.

3.5 Finished product. The finished product shall comply with the following characteristics:

a. There shall be no foreign material such as, but not limited to, dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, moldy, or caramelized.

c. Type III shall possess characteristic flavor, as applicable.

d. Types I through III shall contain not more than 6.0 percent moisture and type IV shall contain not more than 5.0 percent moisture.

e. Types I through III shall contain not more than 1.6 percent salt on an "as is" basis.

f. All types shall be of such firmness that the bars can be easily eaten in the dry form or prepared as a moist cereal by crumbling into a dish, adding water or milk and stirring.

g. Style A bar shall weigh not less than 2 ounces (57g) and style B bar shall weigh not less than 1.5 ounces (43g).

3.5.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Para 110, "Current Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

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3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard method described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Envelope and wrapper examination. The envelopes and wrappers shall be examined for the defects listed in table I. The lot size shall be expressed in bars. The sample unit shall be one filled and sealed envelope or wrapper. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major and 6.5 for minor defects.

TABLE I. Envelope and wrapper defects

Category		Defect
Major	Minor	Envelope
101		Tear, hole, or open seal.
102		Inside dimensions or type, not as specified (see 5.1.1.1). 1/
103		Foreign odor.
104		Excessive air in envelope.
105		Nomenclature missing, incorrect or illegible.
106		Envelope not formed in manner specified.
107		Not clean.
	201	Color not as specified.
	202	Labeling information other than nomenclature missing, incorrect or illegible.
		Plastic Wrapper
108		Not a cellophane or polypropylene wrap
	203	Not close fitting.
109		Does not completely cover product.

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TABLE I. Envelope and wrapper defects (cont'd)

Category		Defect
Major	Minor	Plastic Wrapper
	204	Not closed by one of methods specified.
110		Tear, hole or open seal.
111		Nomenclature missing, incorrect or illegible.
	205	Other required label information missing, incorrect or illegible.
112		Not clean.

1/ Report to the nearest 1/16 inch.

4.5.4 Envelope leakage and delamination examination. The envelopes shall be examined for the defects listed in table II. The lot size shall be expressed in envelopes. The sample unit shall be one filled and sealed envelope. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE II. Envelope leakage (see 4.6.5) delamination defects

Category	Defect
Major	
101	Leakage.
102	Delamination. 1/ 2/

1/ Examine envelope after removal from water.

2/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or farther from food product edge of seal. Envelopes exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be

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rapidly flexed by rotating both hands in alternating clockwise-counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the envelope material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be scored as a defect.

4.5.5 Net weight examination. The filled and sealed envelopes or wrappers shall be examined for the defects listed in table III. The lot size shall be expressed in bars. The sample unit shall be one filled and sealed envelope wrapper. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for minor defects.

TABLE III. Net weight defects

Category	Defect
Minor	
201	Style A bar weighs less than 2.0 ounces or style B bar weighs less than 1.5 ounces. 1/

1/ Report results to nearest 0.1 ounce (nearest 1g). Lot shall be rejected if sample average data indicates lot average net weight is less than specified net weight.

4.5.6 Product examination. The product shall be examined for the defects listed in table IV. The lot size shall be expressed in bars. The sample unit shall be one filled and sealed envelope or wrapper. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major and 6.5 for minor defects.

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TABLE IV. Product defects 1/

Category		Defect
Major	Minor	
101		Cannot be prepared as a moist cereal by crumbling into a dish, adding water or milk, and stirring.
102		Crumbled bar.
103		Not type or style (dimensions) specified. 2/
104		Too hard to bite in the dry form.
105		Does not possess characteristic flavor, as applicable (type III product).
	201	Corners not rounded (style A only). 3/

1/ Presence of foreign material (e.g. glass, dirt, insect part, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Report to the nearest 1/16 inch.

3/ Fit the bar into a square corner. A defect shall be scored if the rounded corner of the bar (tangent) is closer than 1/4 inch (reported to the nearest 1/16 inch) to the corner of the square.

4.5.7 Product testing. The finished product shall be tested for moisture and salt for conformance to the requirements in 3.5 and for vitamins for types I, II, III and IV only in the fortified product for conformance to the requirements of 3.3. Procedures for testing shall be in accordance with 4.6. Lot size shall be expressed in terms of cases. The sample for testing for vitamins (when required), and salt shall be approximately 1 pound of packaged bars derived equally from the number of cases indicated by inspection level S-2. The bars shall be composited just prior to testing. Test results shall be reported to the nearest unit or decimal place as specified for the pertinent requirements except that the results for vitamin A palmitate shall be reported to the nearest 50 I.U. The lot shall be rejected if test results indicate nonconformance to requirements. The sample unit for testing for moisture shall be one packaged bar from one case, and the sample size shall be the number of bars indicated by inspection level S-2. The AQL, expressed as percent defective, shall be 1.5. Results for moisture shall be reported to the nearest 0.1 percent and more than 5.0 or 6.0 percent moisture as applicable in a sample unit shall be a defect.

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4.5.8 Examination of shipping containers. The shipping containers shall be examined in accordance with PPP-B-636 for the defects listed in table V. The lot size shall be expressed in shipping containers. The sample unit shall be one filled shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for minor defects.

TABLE V. Shipping container defects

Category		Defect
Major	Minor	
101		National stock number, item description, contract number, or date of pack markings incorrect, missing, or illegible.
	201	Other required markings missing, incorrect, or illegible.
	202	More than 30 pounds packed in box.
	203	Product not packed in layers.
102		More than one type of product in container.
103		Cushioning material missing or not as specified.
104		Polyethylene liner not used with plastic wrapped products.
105		Liner missing or not as specified, when applicable.

4.6 Methods of inspection.

4.6.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists:

Finished product

Test		Source	Method
Moisture	Chapter:	Sugars and sugar products	Moisture, vacuum drying
Salt	Chapter:	Eggs and egg products	Chloride in eggs (method II)

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4.6.2 Shortening, hydrogenated. Hydrogenated shortening shall be tested for stability (AOM) in accordance with Official and Tentative Methods of the American Oil Chemists Society, Chapter: Commercial Fats and Oils, Method Cd 12-57 to determine compliance with the requirements of 3.2.4.

4.6.3 Vitamin determination. The vitamins shall be determined by Methods of Vitamin Assay prepared and edited by the Association of Vitamin Chemists, Inc., 3rd Ed. (Vitamin A, p. 70-79; thiamine hydrochloride p. 127-140, ascorbic acid, p. 299-306, p. 317-327; pyridoxine hydrochloride p. 212-219).

4.6.4 Colorimetric test for tyrosinase (catechol oxidase).

a. Prepare a 5 percent weight/volume aqueous solution of catechol in distilled or deionized water.

b. Select approximately 1 pound of fresh green oat groats to be used as a standard. Split the sample into two portions and autoclave one portion of the groats at 15 psi steam pressure 250°F for 16 hours to inactivate all enzyme activity. Dry these treated groats and the sample to be tested to the approximate moisture level corresponding to that of the untreated groats, approximately 11.5 percent moisture.

c. Grind the heat treated sample to be tested, and then the untreated groat samples through a medium screen in a Wiley mill in this order. Keep the three samples separate.

d. Weight 5 grams of each sample and place in individual 250 mL Erlenmeyer flasks with 100 mL of distilled or deionized water.

e. Mix well, stopper the flasks, and incubate for 1 hour at 37°C in a water bath.

f. Agitate each flask once and filter the contents through a Whatman #1 filter paper. The aqueous extract can be centrifuged first to facilitate filtering if this is necessary. Discard the first 5-10 mL of filtrate and collect the remainder in a 125 mL Erlenmeyer flask and save for assay.

g. Pipette a 20 mL aliquot of each of the samples into 50 mL Erlenmeyer flasks and add 1 mL of the 5 percent catechol solution to each, mix well, and allow to stand for 15 minutes.

h. The green groats sample will develop a positive brownish-yellow to pink-brown color. The standard control (heat treated) should be nearly colorless. The oat sample for incorporation into the granola mix should also be nearly colorless. The presence of a brownish-yellow to pink-brown color shall be cause for rejection of the lot.

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4.6.5 Leakage examination. The filled and sealed envelope shall be examined for leakage by submerging in water, contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered a sign of leakage.

4.6.6 Coconut fat. Coconut fat shall be tested in accordance with Official and Tentative Methods of the American Oil Chemists Society, Chapter: Commercial Fats and Oils, Method Ca 2B-38 for moisture and impurities, Method Cc 2-38 for Wiley Melting Point, Method Cd 10-57 for solid fat index, Method Cd 8-53 for peroxide value, Method Ca 5A-40 for free fatty acids, Method Cd 1-25 for Iodine Value and Method Cc 13B-45 for color.

5. PACKAGING

5.1 Preservation. Preservation shall be level A. The product shall be unit packed in accordance with 5.1.1.1 or 5.1.1.2, as specified (see 6.1 and 6.2).

5.1.1 Envelopes. Each bar shall be unit packed individually in an envelope formed from one sheet of material folded on edge and heat sealed on the remaining three edges. The inside dimensions of the unfilled envelope for style A bar shall be 2-3/4 inches in width (inside seal edge to inside seal edge) and 5 inches in length (inside bottom edge to opening), for style B bar shall be 2-1/8 inches in width (inside seal edge) and 5 inches in length (inside bottom edge to opening), ± 1/16 inch in either direction. Care shall be taken to expel all excess air by hand, by evacuation, or by gentle mechanical (roller) techniques, prior to sealing the bag. The envelope material shall be made by hot extruding polyethylene to 0.00035-inch thick aluminum foil, and laminating the plain side of the foil to 0.0005-inch thick polyester. The polyethylene shall have a minimum thickness of 0.001 inch. The adhesive shall be an odorless and permanently plastic water-resistant material other than wax. The combined sheet shall show no evidence of delamination when made into envelopes and heat sealed. The polyethylene coated side of the material shall comprise the interior surface of the envelope. The exterior surface of the bag shall be colored overall with a color in the number range from 34052 to 34108, inclusive, of FED-STD-595. The filled and sealed envelope shall not show leakage when examined in accordance with 4.6.5.

5.1.2 Plastic wrap. Each bar of the product shall be wrapped and heat sealed in either a sheet of 0.0014-inch thick cellophane complying with type IV, class 2 of L-C-110 or a minimum 0.001-inch thick oriented heat sealable polypropylene. The wrapper shall be close forming and shall completely cover the product. The wrapper shall be completely sealed, except that when the wrapper is folded so that the two edges are folded under the bottom of the bar and held in place by overlapping the remaining two edges on top of the first two edges, only the bottom area need be heat sealed.

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5.2 Packing. Packing shall be level C.

5.2.1 Level C (for shipment to ration assembler). Not more than 30 pounds of product preserved as specified in 5.1 shall be packed flat in a fiberboard box constructed and closed in accordance with style RSC-L, class domestic, variety SW, grade 200 of PPP-B-636. Cushioning material not less than 1/8-inch thick conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16-inch thick conforming to PPP-C-1797, or cushioning material conforming to type 7, class 1 of PPP-C-1752, shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual unit packs. Plastic wrapped bars shall be enclosed in a waterproof case liner, fabricated and closed in accordance with MIL-L-10547 for subsistence items.

5.2.2 Unit loads. When specified (see 6.1), product packed as specified in 5.2.1 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of the load specified. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.3 Labeling and marking.

5.3.1 Envelopes and plastic wraps. The envelopes or plastic wraps shall have the following information clearly and legibly printed in black ink:

CORN FLAKE BAR

or as applicable

RICE-CORN FLAKE BAR

or

FRUIT FLAVORED CORN FLAKE BAR
(- Orange flavor or - Lemon flavor,
as applicable).

GRANOLA BAR

(In letters larger than any other used for the label).

Any other information required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number of date of this document.
b. Type, class (as applicable), and style of product required (see 1.2).
c. When a first article is required (see 3.1, 4.4, and 6.3).
d. Provisions for approved preproduction samples (see 3.5.1 and 6.3).
e. Bar unit packing required (see 5.1.1.1, 5.1.1.2, and 6.2).
f. When vitamin fortification is required (see 3.4).
g. Type and class of unit load when unit loading is required (see 5.2.2).

6.2 Plastic wrapped bars. Bars used in canned rations, such as Food Packet, Survival, General Purpose, MIL-F-43231 shall be plastic wrapped for ease in can filling and closing.

6.3 First article inspection. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Bars
Cereal Bars
Survival

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC, MS
DLA - SS

Preparing activity:

Army - GL

Project No. 8920-0478

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-35074G	Cereal Bars, Survival-Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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